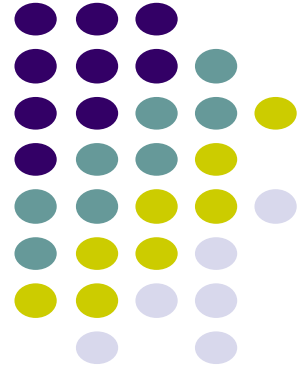


# ErSE222: Machine learning in Geoscience

Feb. 15<sup>th</sup>, 2026

*Tariq Alkhalifah and Omar Saad*





# Backpropagation (BCE)

General algorithm, called **backpropagation** → used to compute gradients of any complex function that be composed of simple, differentiable functions

$$z = \mathbf{x}^T \boldsymbol{\theta}, \quad a = \sigma(z), \quad \mathcal{L} = -(y \log(a) + (1 - y) \log(1 - a))$$



Chain-  
rule

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}} = \frac{\partial \mathcal{L}}{\partial a} \frac{\partial a}{\partial z} \frac{\partial z}{\partial \boldsymbol{\theta}}$$

# Backpropagation (BCE)



Let's now (for once) do the math:

$$\mathcal{L} = -(y \log(a) + (1 - y) \log(1 - a))$$

$$\frac{\partial \mathcal{L}}{\partial a} = -\frac{y}{a} + \frac{1 - y}{1 - a} = \frac{-y(1 - a) + (1 - y)a}{a(1 - a)}$$

$$\frac{\partial a}{\partial z} = a(1 - a) \quad \boxed{a = \sigma(z)}$$

# Backpropagation (BCE)



$$y = \sigma(x) = \frac{1}{1+e^{-x}}$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$u = 1 + e^{-x}$$

$$y = \frac{1}{u}$$

$$\frac{dy}{du} = -\frac{1}{u^2}$$

$$\frac{du}{dx} = -e^{-x}$$

## Backpropagation (BCE)



$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$\frac{dy}{du} = -\frac{1}{u^2} \quad \frac{du}{dx} = -e^{-x}$$

$$\frac{dy}{dx} = \frac{e^{-x}}{u^2} \quad u = 1 + e^{-x}$$

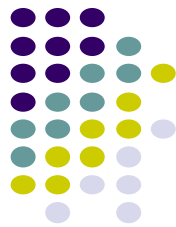
$$\frac{dy}{dx} = \frac{e^{-x}}{(1+e^{-x})^2}$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$1-\sigma(x) = \frac{e^{-x}}{1+e^{-x}}$$

$$\sigma'(x) = \sigma(x) \cdot (1-\sigma(x))$$

# Backpropagation (BCE)



Let's now (for once) do the math:

$$z = \mathbf{x}^T \boldsymbol{\theta}, \quad a = \sigma(z), \quad \mathcal{L} = -(y \log(a) + (1 - y) \log(1 - a))$$

$$\frac{\partial \mathcal{L}}{\partial a} = -\frac{y}{a} + \frac{1 - y}{1 - a} = \frac{-y(1 - a) + (1 - y)a}{a(1 - a)}$$

$$\frac{\partial a}{\partial z} = a(1 - a) \quad \boxed{a = \sigma(z)}$$



$$\frac{\partial \mathcal{L}}{\partial z} = \frac{\partial \mathcal{L}}{\partial a} \frac{\partial a}{\partial z} = -y(1 - a) + (1 - y)a = a - y = dz$$

# Backpropagation (BCE)



$$z = \mathbf{x}^T \boldsymbol{\theta}, \quad a = \sigma(z), \quad \mathcal{L} = -(y \log(a) + (1 - y) \log(1 - a))$$

Let's now (for once) do the math:

$$\frac{\partial \mathcal{L}}{\partial z} = \frac{\partial \mathcal{L}}{\partial a} \frac{\partial a}{\partial z} = -y(1 - a) + (1 - y)a = a - y = dz$$

$$\frac{\partial z}{\partial w_i} = x_i, \quad \frac{\partial z}{\partial b} = 1 \quad \frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}} = \frac{\partial \mathcal{L}}{\partial a} \frac{\partial a}{\partial z} \frac{\partial z}{\partial \boldsymbol{\theta}}$$



$$\frac{\partial \mathcal{L}}{\partial w_i} = dz \cdot x_i = dw_i, \quad \frac{\partial \mathcal{L}}{\partial b} = dz = db$$

# Backpropagation (BCE)



$$\mathbf{z} = \mathbf{X}_{train}^T \boldsymbol{\theta}$$

$$\mathbf{a} = \sigma(\mathbf{z})$$

$$\mathbf{dz} = \mathbf{a} - \mathbf{y}$$

$$\mathbf{dw} = \frac{1}{N_s} \mathbf{X}_{train} \mathbf{dz}$$

$$db = \frac{1}{N_s} \mathbf{1}^T \mathbf{dz}$$

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha \mathbf{dw}$$

$$b \leftarrow b - \alpha db$$

} Forward

} Backward

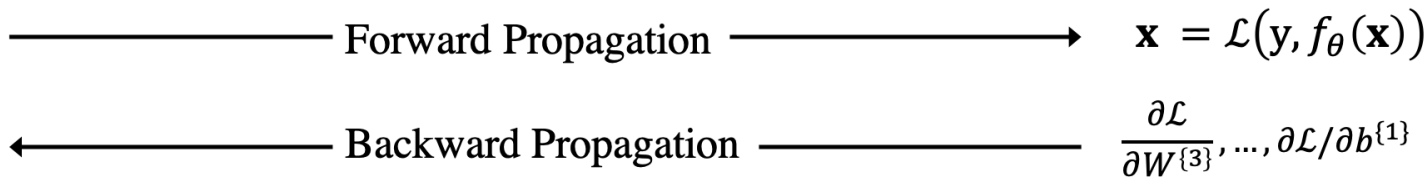
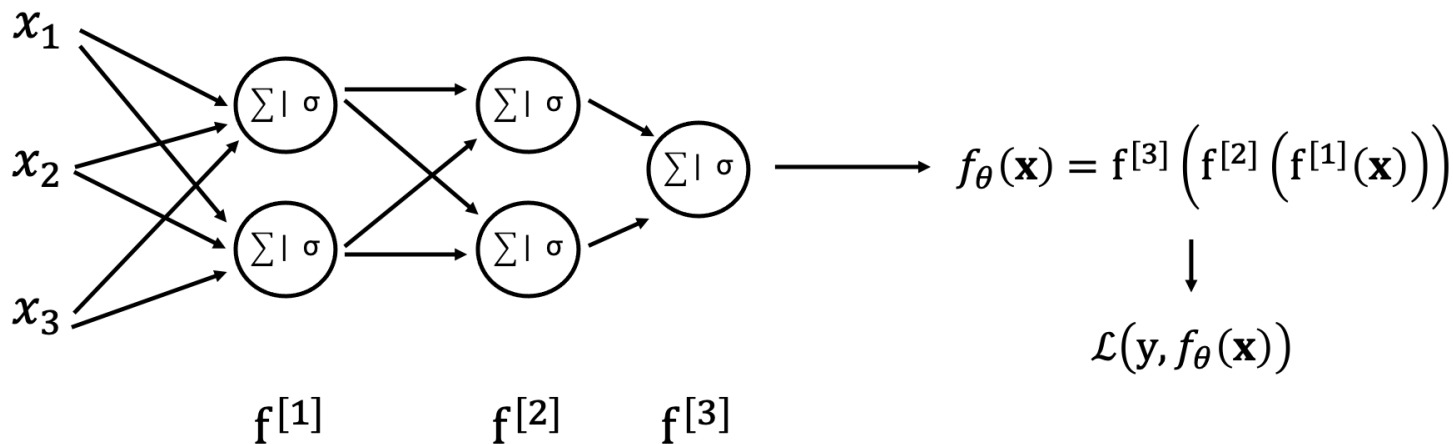
} Update



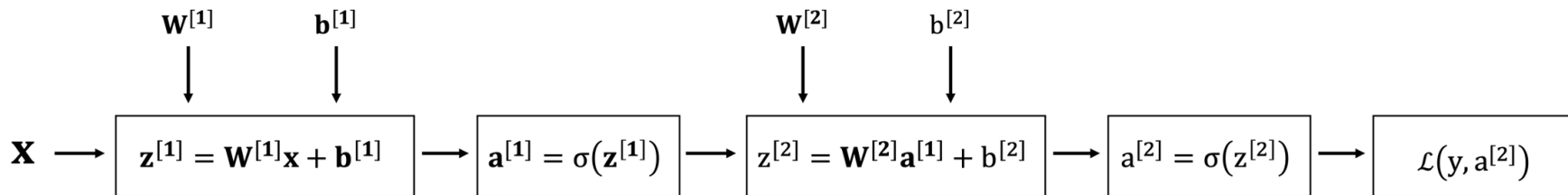


# MLP

# Backpropagation (BCE)



# Backpropagation (BCE)



And write in reversed order (evaluated right to left):

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{[1]}} = \frac{\partial \mathbf{z}^{[1]}}{\partial \mathbf{W}^{[1]}} \frac{\partial \mathbf{a}^{[1]}}{\partial \mathbf{z}^{[1]}} \frac{\partial \mathbf{z}^{[2]}}{\partial \mathbf{a}^{[1]}} \frac{\partial \mathbf{a}^{[2]}}{\partial \mathbf{z}^{[2]}} \frac{\partial \mathcal{L}}{\partial \mathbf{a}^{[2]}}$$



Effective order of operations

# Backpropagation (BCE)

$$\mathbf{z} = \mathbf{X}_{train}^T \boldsymbol{\theta}$$

$$\mathbf{a} = \sigma(\mathbf{z})$$

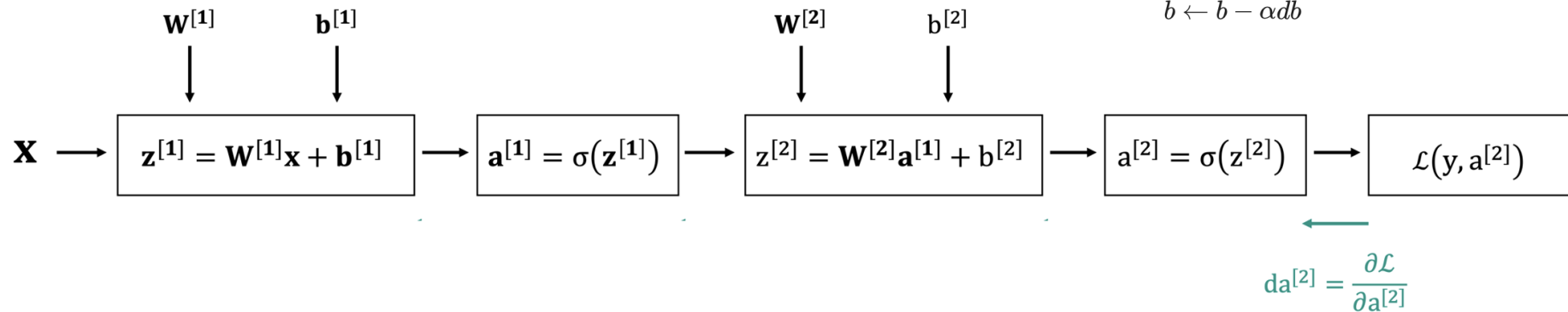
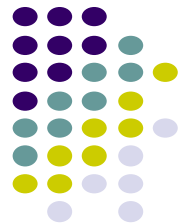
$$\mathbf{dz} = \mathbf{a} - \mathbf{y}$$

$$d\mathbf{w} = \frac{1}{N_s} \mathbf{X}_{train} d\mathbf{z}$$

$$db = \frac{1}{N_s} \mathbf{1}^T d\mathbf{z}$$

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha d\mathbf{w}$$

$$b \leftarrow b - \alpha db$$



# Backpropagation (BCE)



$$\mathbf{z} = \mathbf{X}_{train}^T \boldsymbol{\theta}$$

$$\mathbf{a} = \sigma(\mathbf{z})$$

$$\mathbf{dz} = \mathbf{a} - \mathbf{y}$$

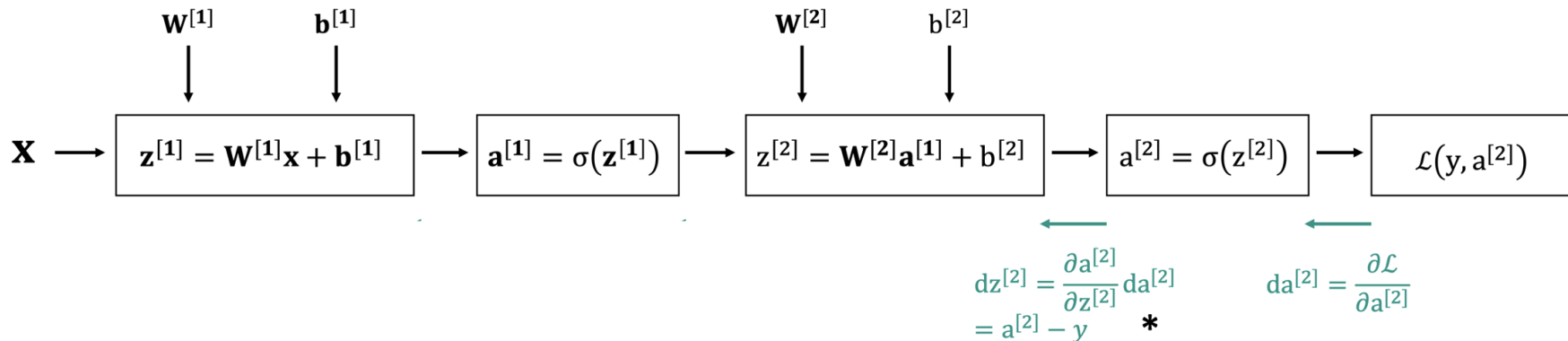
$$\mathbf{dw} = \frac{1}{N_s} \mathbf{X}_{train} \mathbf{dz}$$

$$db = \frac{1}{N_s} \mathbf{1}^T \mathbf{dz}$$

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha \mathbf{dw}$$

$$b \leftarrow b - \alpha db$$

$$\frac{\partial \mathcal{L}}{\partial z} = \frac{\partial \mathcal{L}}{\partial a} \frac{\partial a}{\partial z} = -y(1-a) + (1-y)a = a - y = dz$$



\* Assuming  $\mathcal{L}$ =BCE

# Backpropagation (BCE)



$$\mathbf{z} = \mathbf{X}_{train}^T \boldsymbol{\theta}$$

$$\mathbf{a} = \sigma(\mathbf{z})$$

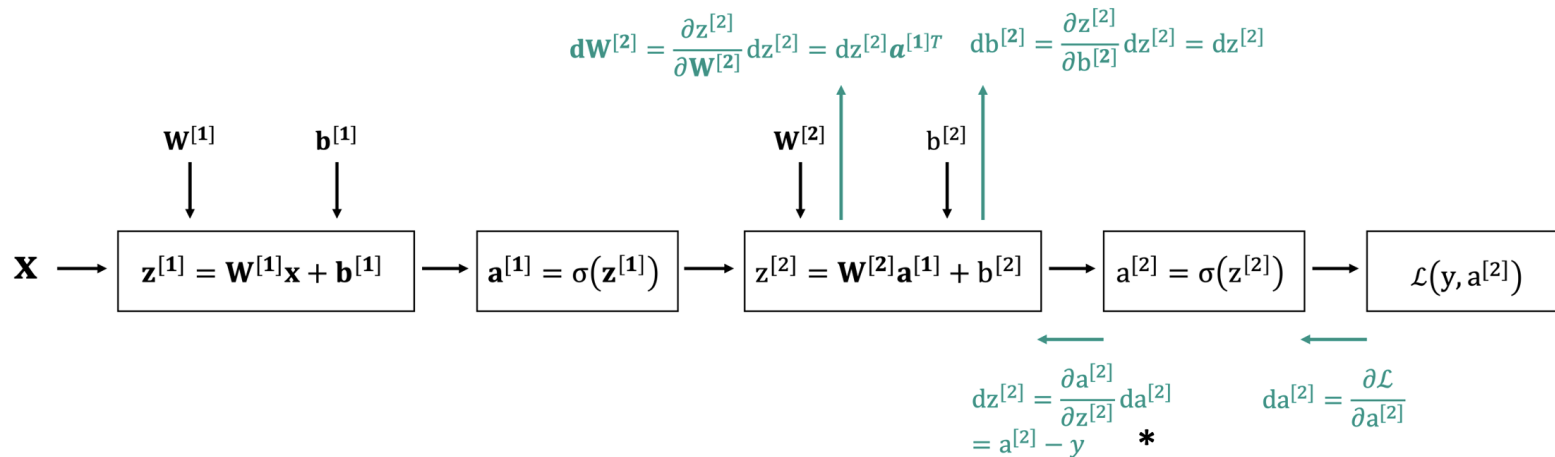
$$\mathbf{dz} = \mathbf{a} - \mathbf{y}$$

$$\mathbf{dw} = \frac{1}{N_s} \mathbf{X}_{train} \mathbf{dz}$$

$$db = \frac{1}{N_s} \mathbf{1}^T \mathbf{dz}$$

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha \mathbf{dw}$$

$$\frac{\partial \mathcal{L}}{\partial w_i} = dz \cdot x_i = dw_i, \quad \frac{\partial \mathcal{L}}{\partial b} = dz = db$$



\* Assuming  $\mathcal{L}=\text{BCE}$

# Backpropagation (BCE)

$$\frac{\partial \mathcal{L}}{\partial w_i} = dz \cdot x_i = dw_i, \quad \frac{\partial \mathcal{L}}{\partial b} = dz = db$$

$$\mathbf{z} = \mathbf{X}_{train}^T \boldsymbol{\theta}$$

$$\mathbf{a} = \sigma(\mathbf{z})$$

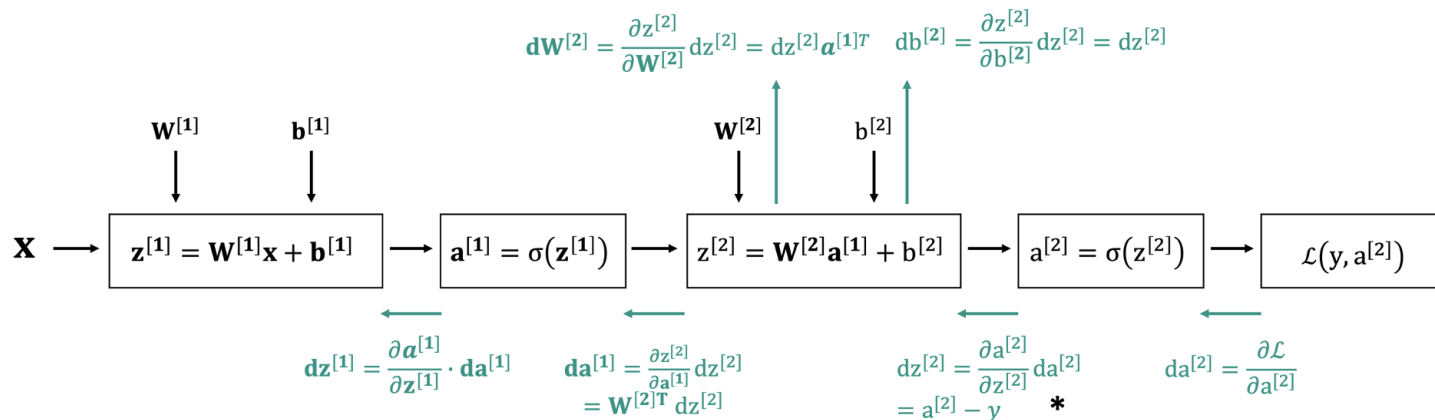
$$\mathbf{dz} = \mathbf{a} - \mathbf{y}$$

$$d\mathbf{w} = \frac{1}{N_s} \mathbf{X}_{train} d\mathbf{z}$$

$$db = \frac{1}{N_s} \mathbf{1}^T d\mathbf{z}$$

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha d\mathbf{w}$$

$$b \leftarrow b - \alpha db$$



\* Assuming  $\mathcal{L}$ =BCE

# Backpropagation (BCE)



$$\mathbf{z} = \mathbf{X}_{train}^T \boldsymbol{\theta}$$

$$\mathbf{a} = \sigma(\mathbf{z})$$

$$\mathbf{dz} = \mathbf{a} - \mathbf{y}$$

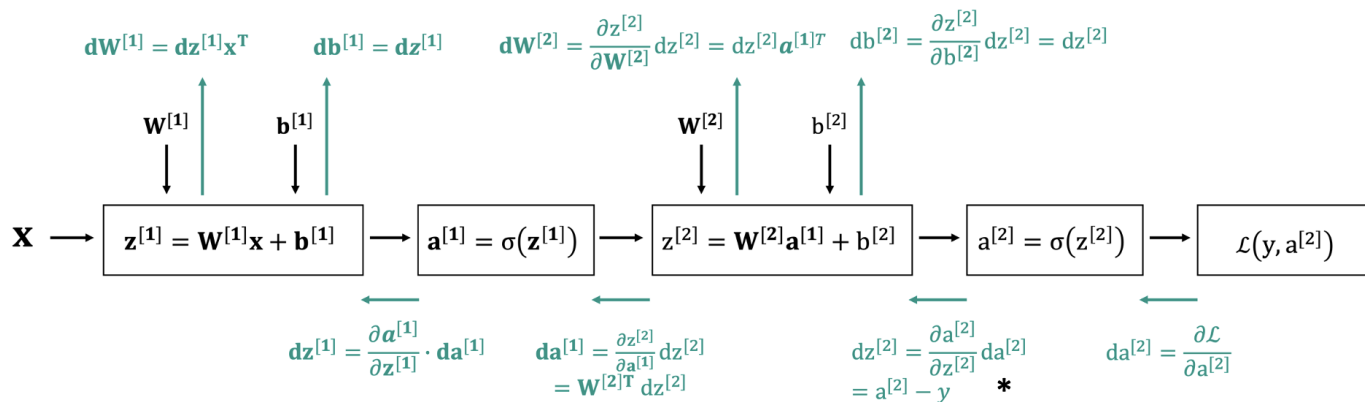
$$d\mathbf{w} = \frac{1}{N_s} \mathbf{X}_{train} d\mathbf{z}$$

$$db = \frac{1}{N_s} \mathbf{1}^T d\mathbf{z}$$

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha d\mathbf{w}$$

$$b \leftarrow b - \alpha db$$

$$\frac{\partial \mathcal{L}}{\partial w_i} = dz \cdot x_i = dw_i, \quad \frac{\partial \mathcal{L}}{\partial b} = dz = db$$



\* Assuming  $\mathcal{L}=\text{BCE}$



# Initialization



- **Random initialization:** general idea

$$w_{i,j}^{[.]} \sim \mathcal{N}(0, 0.01)$$

Not too small: to avoid slow training  
Not too high: to avoid unstable training

$$b_i^{[.]} = 0 \quad (\text{or constant - e.g. for ReLU})$$

# Initialization



- **Random initialization:** more appropriate choices (depend on #units in layer)

- Uniform distributior  $w_{ij}^{[k]} \sim \mathcal{U}(-1/\sqrt{N^{[k]}}, 1/\sqrt{N^{[k]}})$
- Xavier initialization  $w_{ij}^{[k]} \sim \mathcal{N}(0, 1/N^{[k]})$
- He initialization  $w_{ij}^{[k]} \sim \mathcal{N}(0, 2/N^{[k]})$

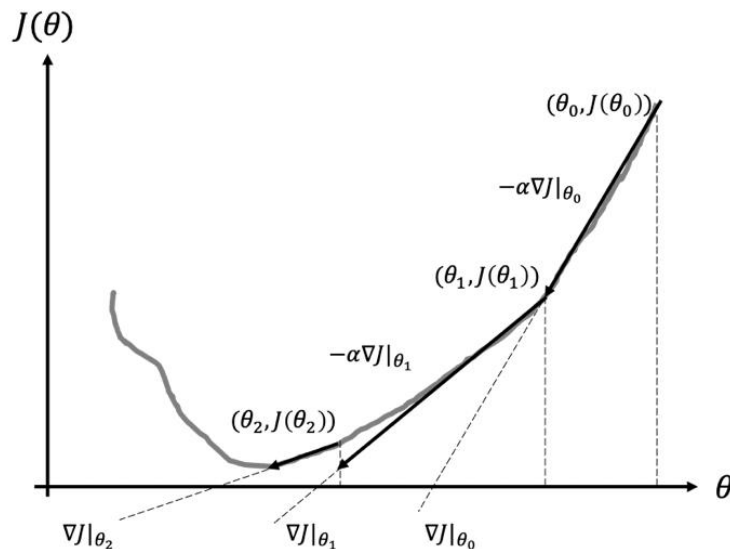


# Optimization

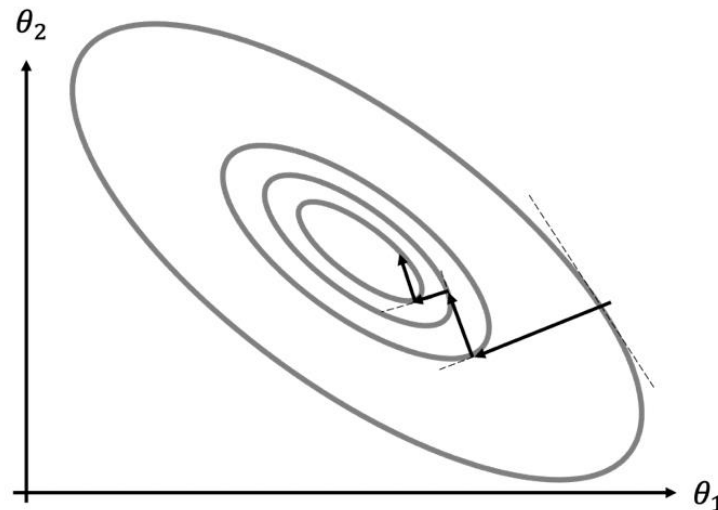
# Gradient-based optimization



1d function



2d function



Use local information of  $f(x)$  to find its minimum/maximum

# Gradient-based optimization



## Gradient descent algorithm:

Initialization: choose  $\theta \in \mathbb{R}$

For  $i = 0, \dots, N - 1$ ;

1. Compute update direction  $d_i = -\nabla J|_{\theta_i}$
2. Estimate step-length  $\alpha_i$
3. Update  $\theta_{i+1} = \theta_i + \alpha_i d_i$

Choice of step-length:

- Constant or with some schedule (in DL)
- Exact
- Backtracking

Extension to n-dimensional problems:

In DL applications,  $\theta = [\theta_1, \theta_2, \dots, \theta_N]$

$$\nabla J = [\delta J / \delta \theta_1, \delta J / \delta \theta_2, \dots, \delta J / \delta \theta_N].$$



# Stochastic gradient descent (SGD)

**Batched gradient descent:**

$$\boldsymbol{\theta}_{i+1} = \boldsymbol{\theta}_i - \alpha_i \nabla J = \boldsymbol{\theta}_i - \frac{\alpha_i}{N_s} \sum_{j=1}^{N_s} \nabla \mathcal{L}_j$$

**Stochastic gradient descent:**

$$\boldsymbol{\theta}_{i+1} = \boldsymbol{\theta}_i - \alpha_i \nabla \mathcal{L}_j$$

**Mini-batch gradient descent:**

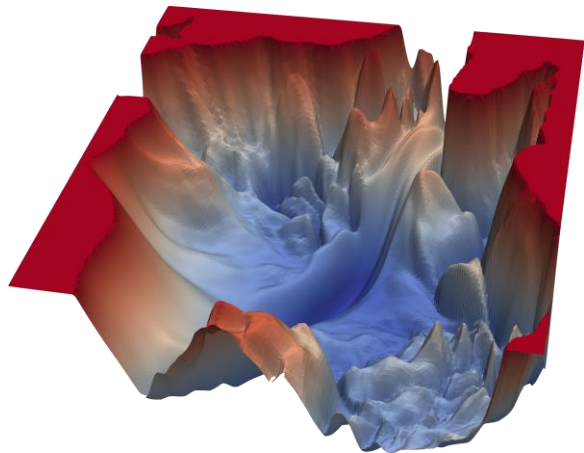
$$\boldsymbol{\theta}_{i+1} = \boldsymbol{\theta}_i - \frac{\alpha_i}{N_b} \sum_{j=1}^{N_b} \nabla \mathcal{L}_j$$

Mini-batch size

# Limitations of SGD



- **Ill-conditioning:** In an ill-conditioned problem, the gradients may change dramatically with small changes in the parameter values. This results in steep gradients in certain directions and shallow gradients in others.
- **Local minima:** the landscape of NN functionals is generally non-convex and populated with a multitude of local minima



## *Model identifiability*

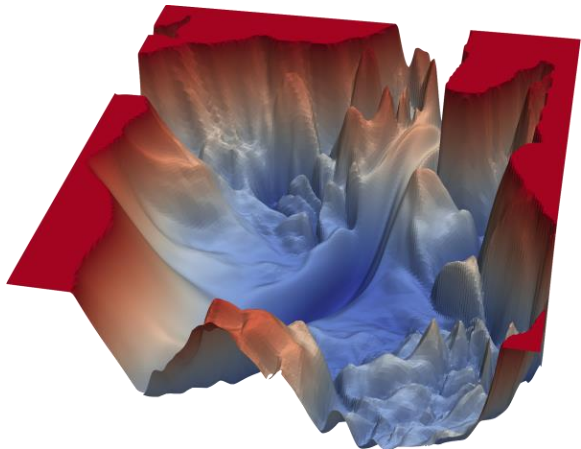
$\theta_{gm}$  Single set of params for optimal performance

$\theta_{gm}, \theta_{lm,1}, \dots$  Multiple models with similar performance

# Limitations of SGD



- **Saddle points (and other flat regions)**: landscapes associated to the training of deep neural networks may have much fewer local minima than we tend to believe... the main hinder to slow training is represented by saddle points



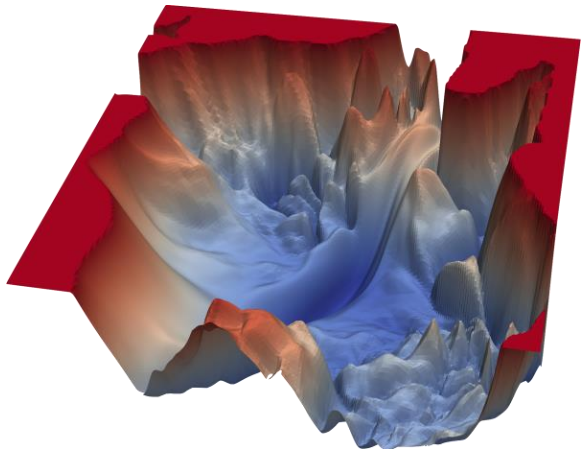
$J(\theta) \rightarrow 0$       Vanishing gradients



# Limitations of SGD



- **Cliffs:** steep regions in the NN landscape with high gradient



$J(\theta) \rightarrow \infty$       Exploding gradients

$\nabla J(\theta_i) = \min(\nabla J(\theta_i), th)$       Gradient clipping

# Cooling



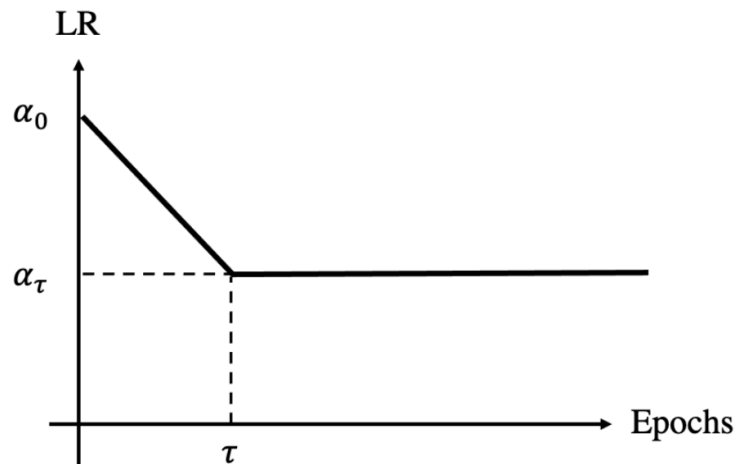
- Linearly decaying learning rate

$$\alpha_i = (1 - \beta)\alpha_0 + \beta\alpha_\tau \quad i < \tau$$

$$\alpha_i = \alpha_\tau \quad i \geq \tau$$

$$\beta = i/\tau$$

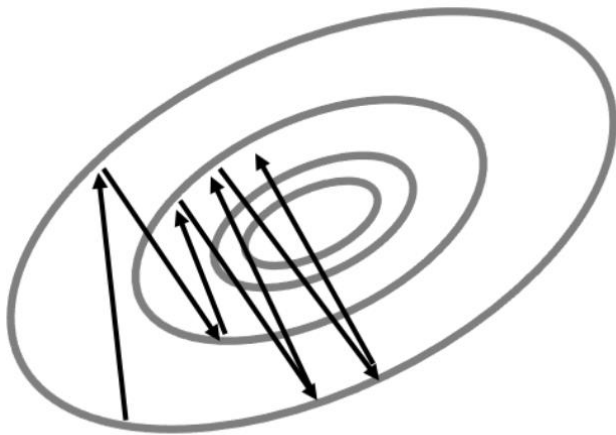
$$\tau \approx 100N_{epochs}, \alpha_\tau = \alpha_0/100$$



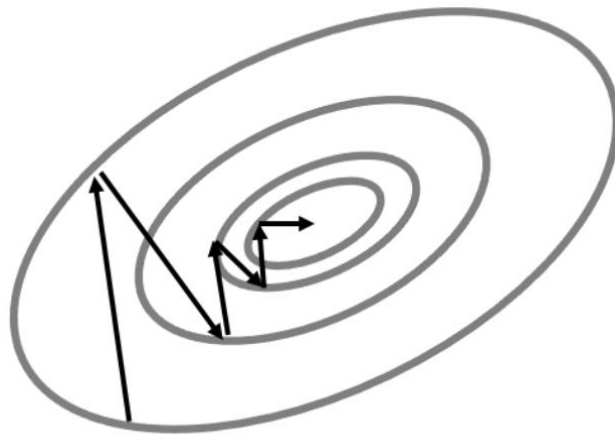
# Cooling



Fixed learning rate



Cooling strategy



# SGD with momentum



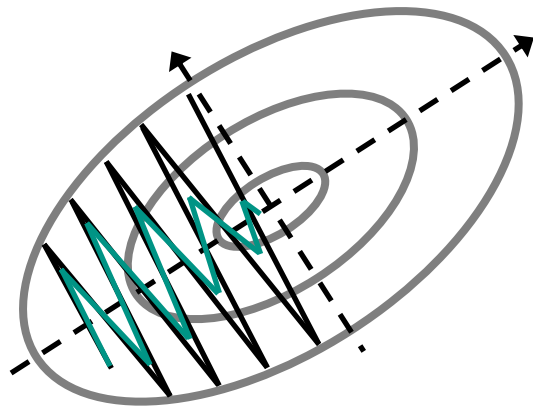
**Idea of momentum** (from Polyak and Nesterov in 60'): use update that is an exponentially decaying moving average of the past gradients.

Velocity                      Momentum term [0,1)

↓                      ↙

$$\mathbf{v}_{i+1} = \gamma \mathbf{v}_i - \mathbf{g}_{i+1} = \gamma \mathbf{v}_i - \frac{\alpha}{N_b} \sum_{j=1}^{N_b} \nabla \mathcal{L}_j$$

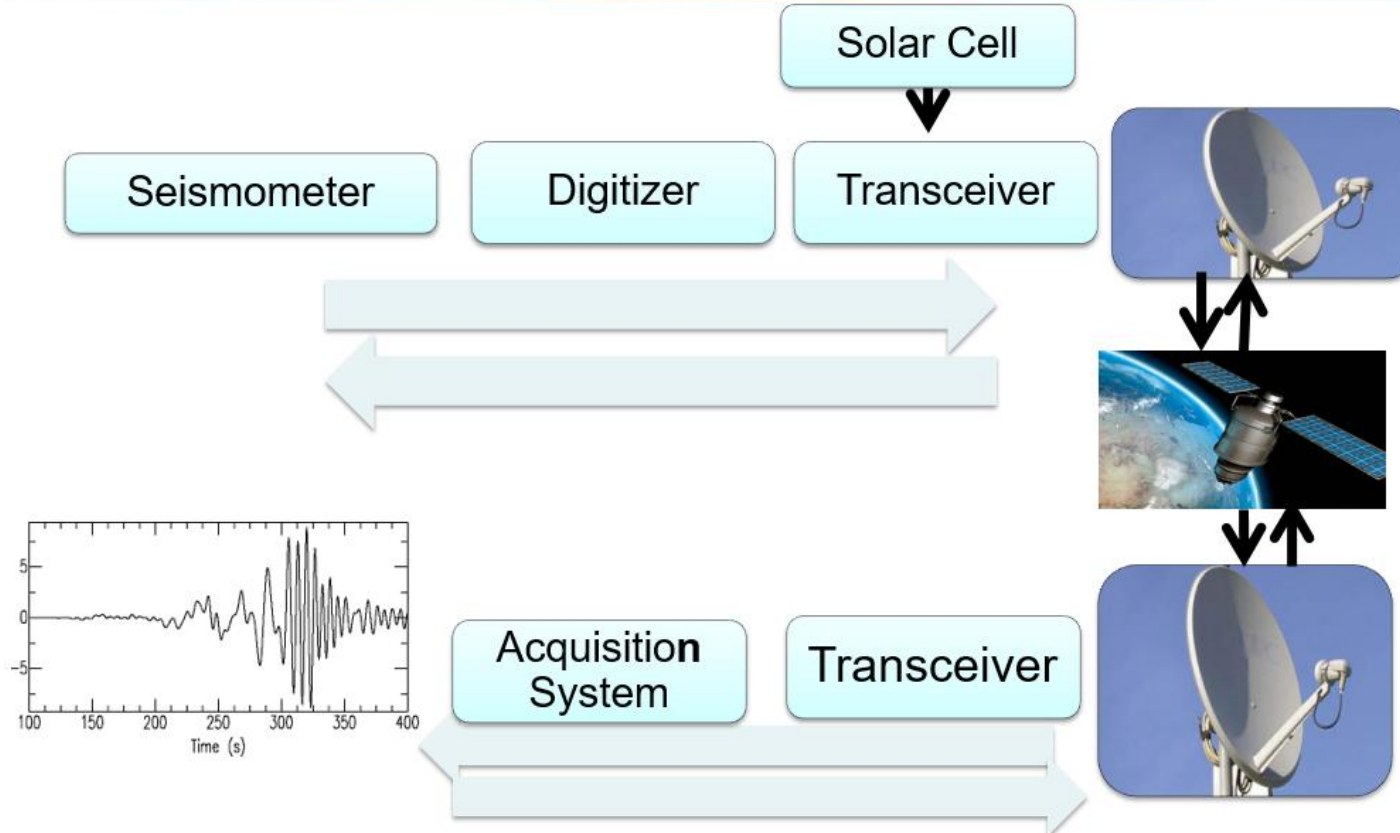
$$\boldsymbol{\theta}_{i+1} = \boldsymbol{\theta}_i + \mathbf{v}_{i+1}$$



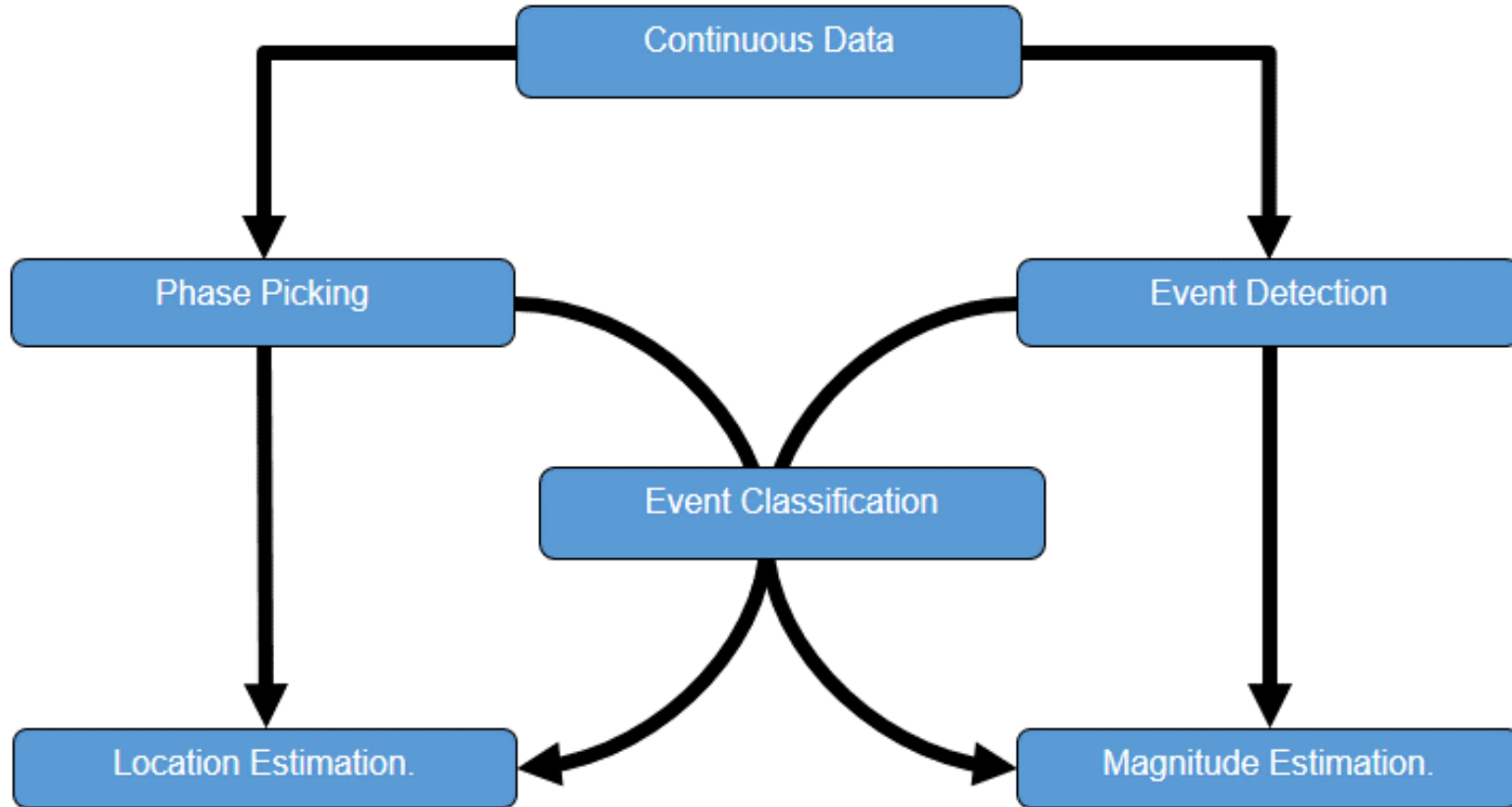


# Application

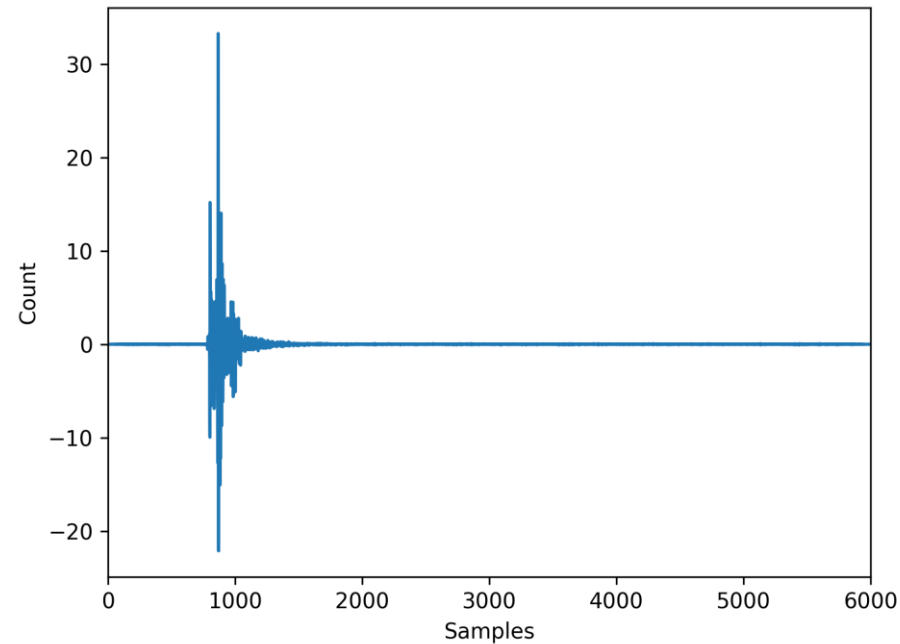
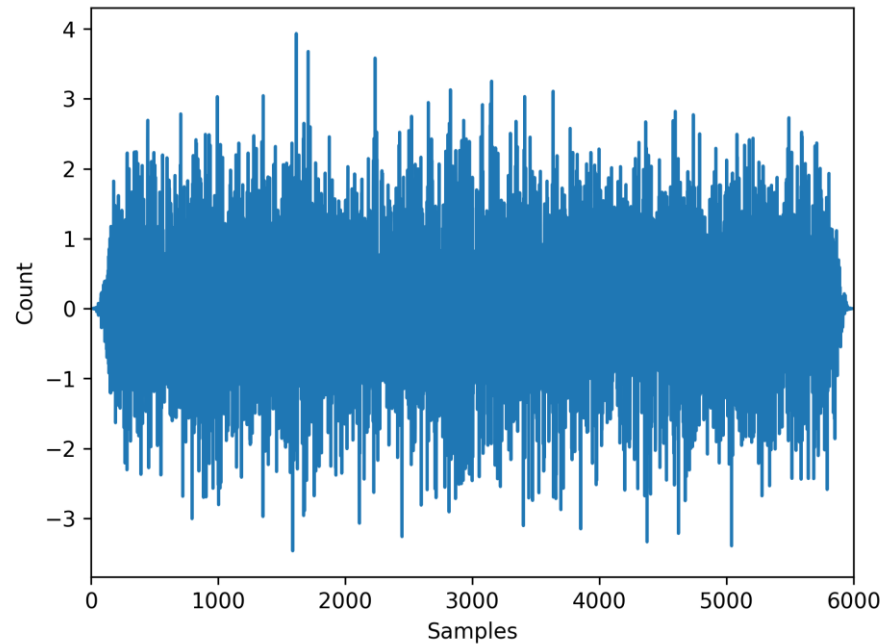
# Earthquake Monitoring System



# Earthquake Monitoring System

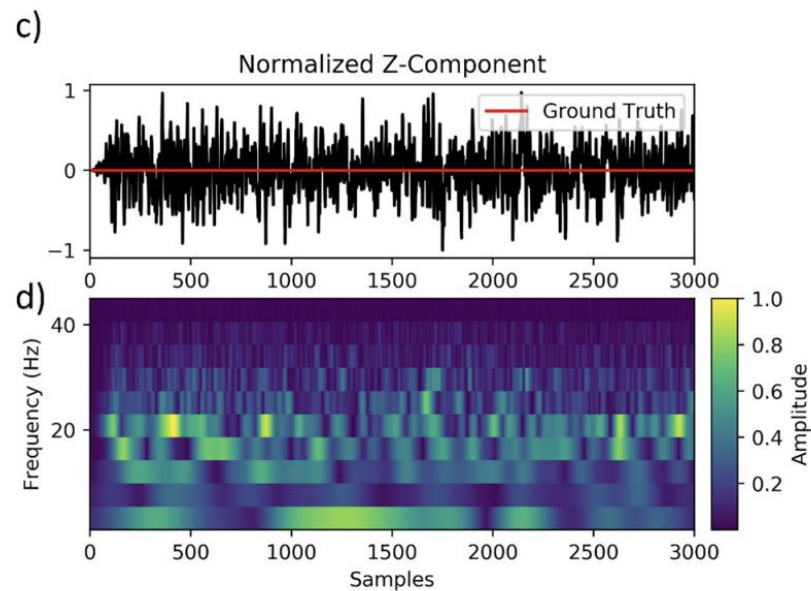
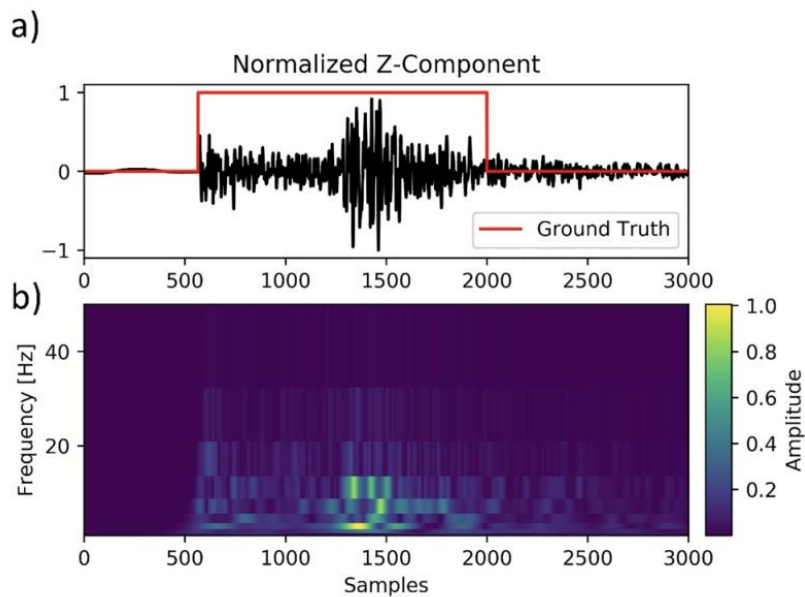


# Seismic Event Classification





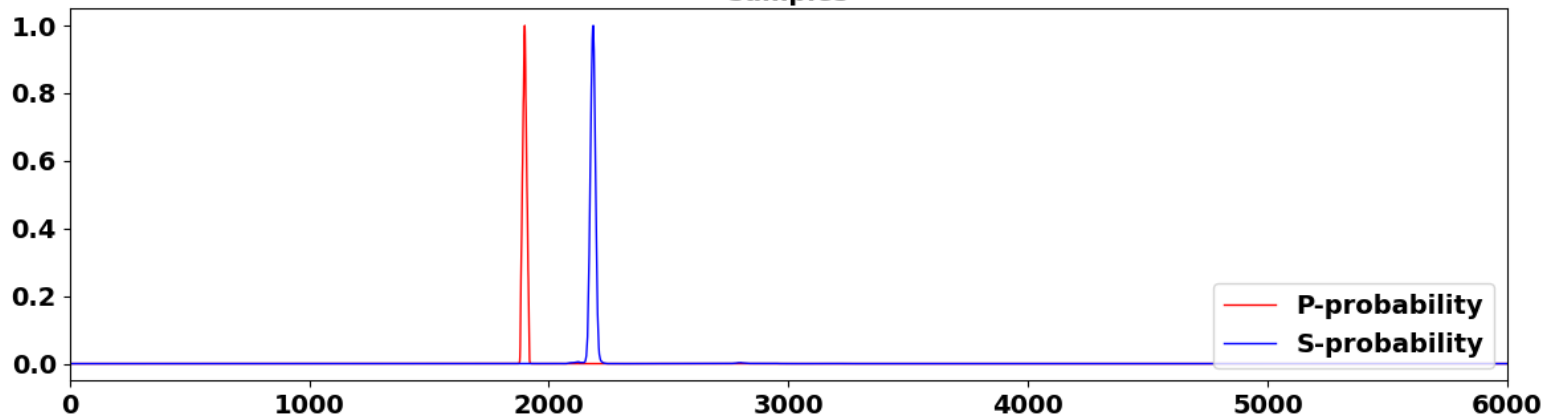
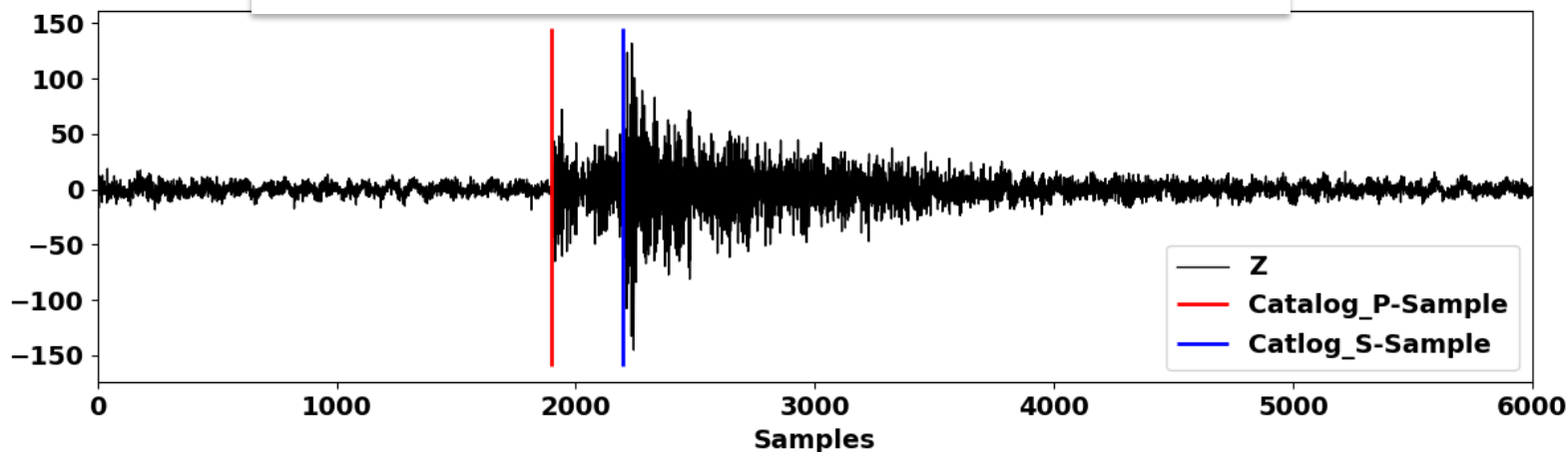
# Sample-based Classification (Detection)



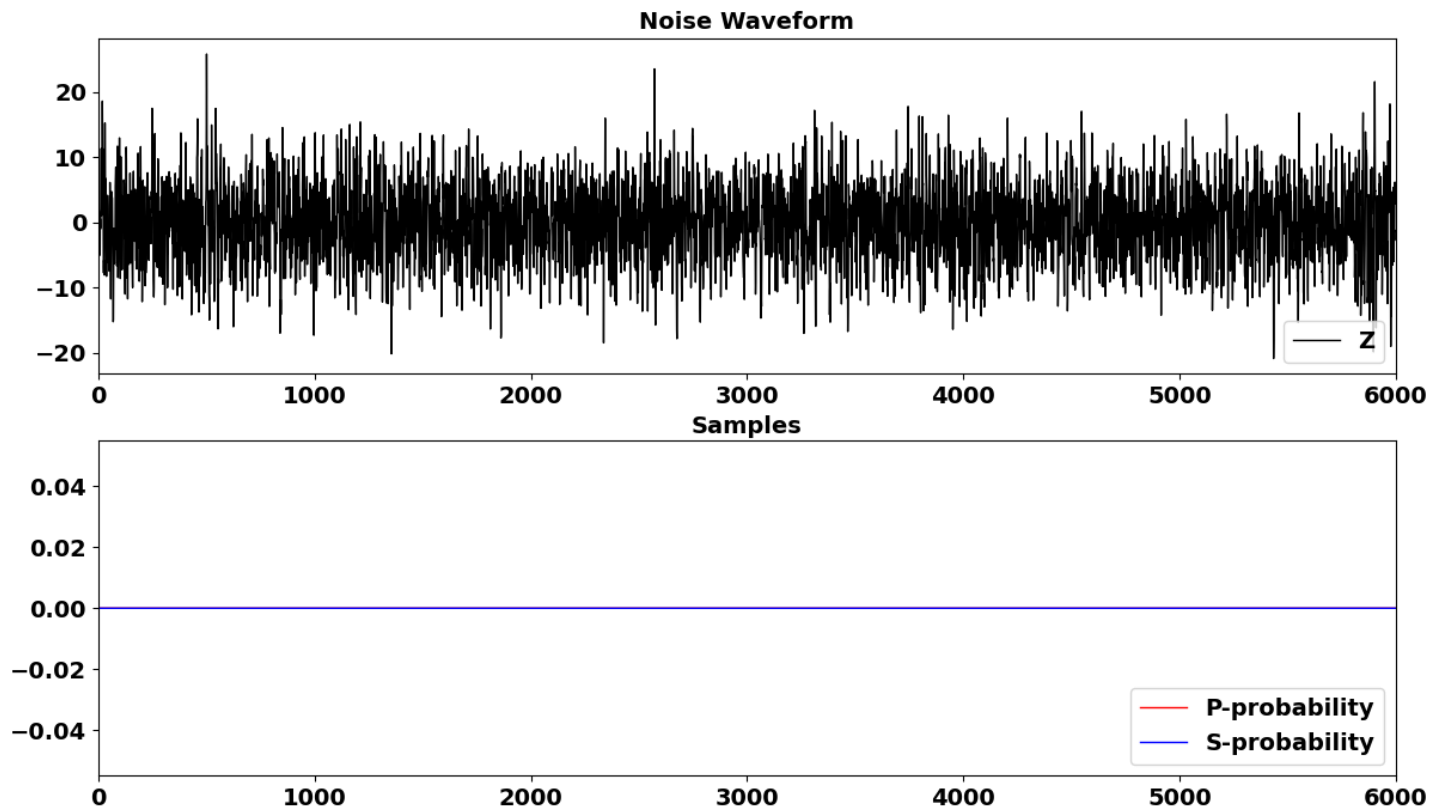
# Sample-based Classification (Picking)



Recorded Earthquake



# Sample-based Classification (Picking)



A wide-angle photograph of a desert landscape. In the foreground, there are prominent, layered rock formations, likely sandstone, with a warm, golden-brown hue. The rocks show clear horizontal strata. Beyond the rocks, a vast, flat valley stretches out, with some distant, low-lying hills and a few small structures visible on the horizon. The sky is a pale, hazy blue. The overall scene is serene and expansive.

***Thank You***

30 17:50